Aimpoint Digital Optimization Coding Challenge

0 Pre-reqs

- Data analysis programming language of your choice
- LP/MIP solver of your choice

1 Instructions

Please submit your solution via email in .txt format (code and results) and a .jpg/.png/etc. picture of your model formulation (can be handwritten, typed etc.). Please label each file appropriately and make sure they are easily identified. Ensure readability and reproducibility; code should be well commented and easy to execute. You will use the dataset provided in the accompanying email. This challenge should take no more than 3 hours. Please focus on understanding/exploring the data as well as answering the client prompts below.

2 Challenge Description

Our client produces four types of cars: electric, compact, intermediate and luxury and two types of heavy vehicles, trucks and vans. Manufacturing limits total production capacity to at most 1,300,000 vehicles per year. Electrics and compacts are built together in a facility with a total annual capacity of 680,000 cars. Intermediate and luxury cars are produced in another facility with capacity of 430,000; and the heavy vehicle facility has a capacity of 315,000. Our client's marketing strategy requires that electrics and compacts must constitute at least half of the product mix for the four car types. Profit margins and market potential are summarized as follows:

Type	Profit / vehicle	Maximum sales (in 1000)
Electric	150	600
Compact	225	400
Intermediate	250	300
Luxury	500	225
Truck	400	225
Van	200	100

According to government regulation, the average fleet fuel efficiency must be at least 30 MPG. Our client would like to use an optimization model to understand the implications of government and corporate policies on its production plans.

Using the cars dataset:

1. What is the heaviest vehicle?
2. What is the car with the highest fuel efficiency?
3. Determine the average fuel efficiency of each type of vehicle

4. Formulate a linear program to determine the maximum profit that can be achieved by our client.
5. Solve the LP using a program of your choice.
6. What is the maximum profit that can be obtained? How many and what percentage of total production will electric cars produced under this plan?
7. Optional Bonus Question:
    a. Unfortunately, due to supply chain and labor constraints, our client has a new requirement. Either luxury cars or trucks can be produced, but not both. Implement this requirement in the model.
    b. What is the new maximum profit that can be obtained? Of luxury cars and trucks, which type of vehicle will be produced?